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Session : Operationalizing IS 18968:2025 to Improve Grid Reliability, Reduce Curtailment, And Enable DER Growth

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Co-Founder, EZ-Q

Grid stress and renewable curtailment:

Signals are already visible;

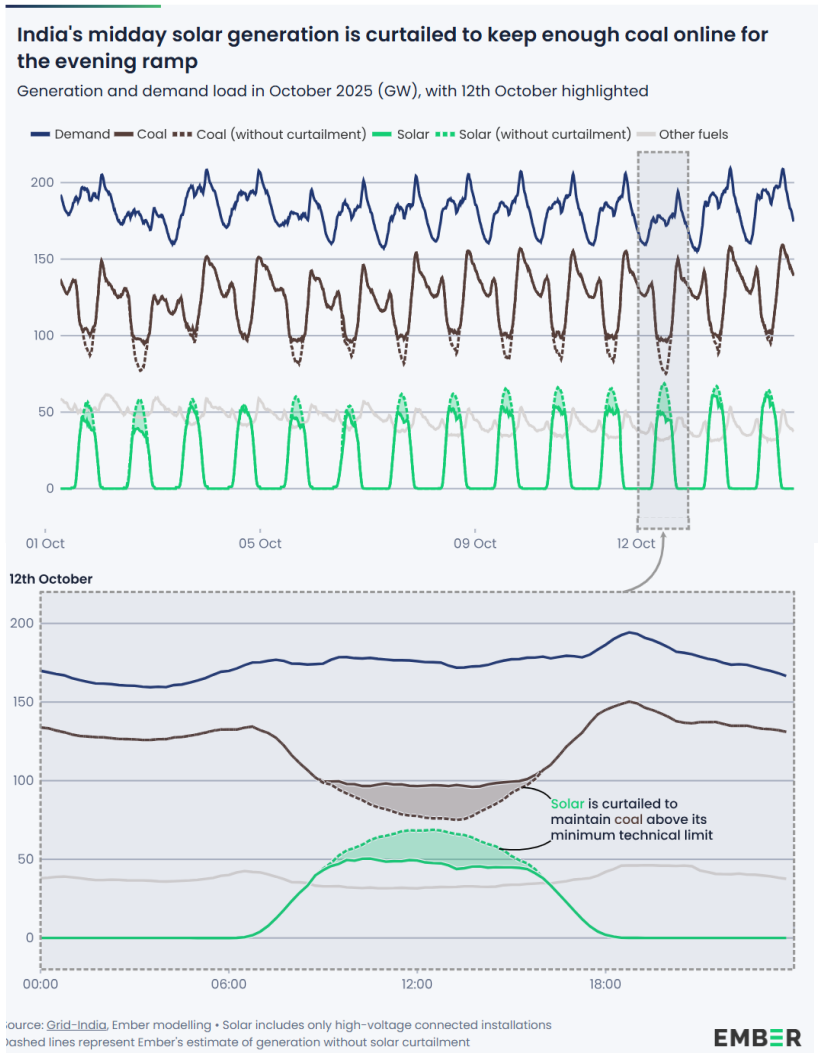
- Curtailment and emergency dispatch reflect congestion,
- Balancing limits, and
- Inadequate grid-support from DER

Recent curtailment indicators (examples)

- May - Dec 2025 :
  - India **curtailed ~2.3 TWh** of solar generation (Ember)
  - Peak-day **RE curtailment >23 GW** reported during this period(GRID-India data)
  - Severe peak-hour curbs reported in Rajasthan (Reuters) —
    - pointing to evacuation and balancing constraints

Underlying operational drivers

- Congestion & evacuation delays
- Low demand / high RE periods → high frequency risk
- Weak-grid pockets (low SCR) at MV pooling points
- Limited feeder voltage regulation and visibility at the edge

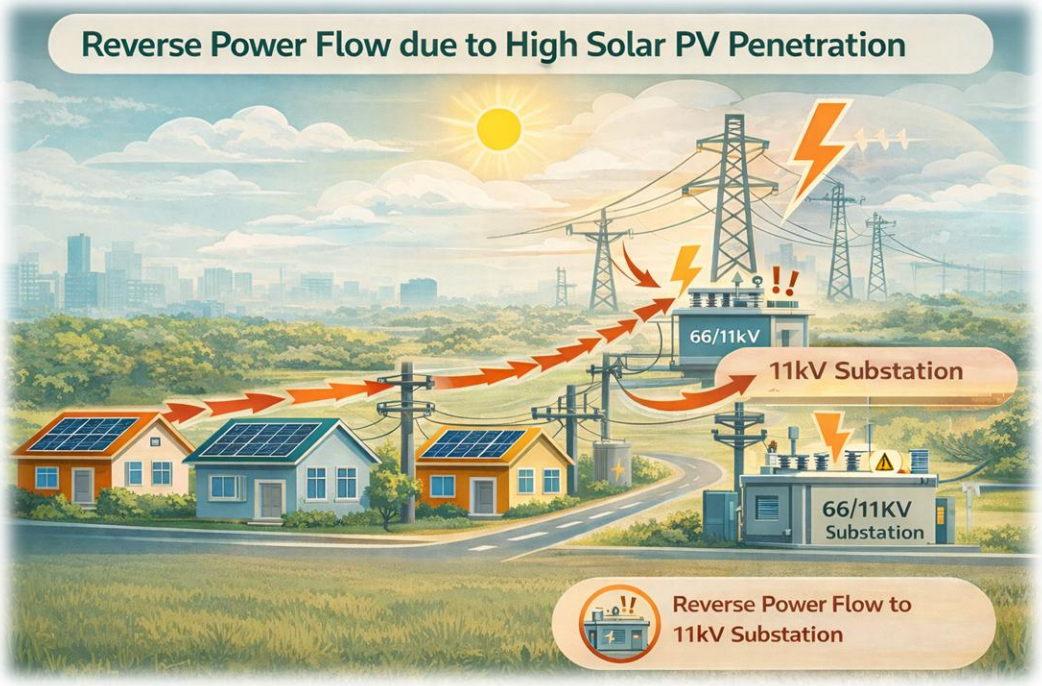


Case Study : DISCOM (India)

Paschim Gujarat Vij Company Limited (PGVCL)

- **One** of the four **government-owned electricity distribution companies** in Gujarat(India)
- **Actively modernizing the network** through underground cabling, advanced system upgrades, and loss-reduction initiatives.
- It has over **300,000 RTS connections**, which is the **highest in any distribution utility in India**

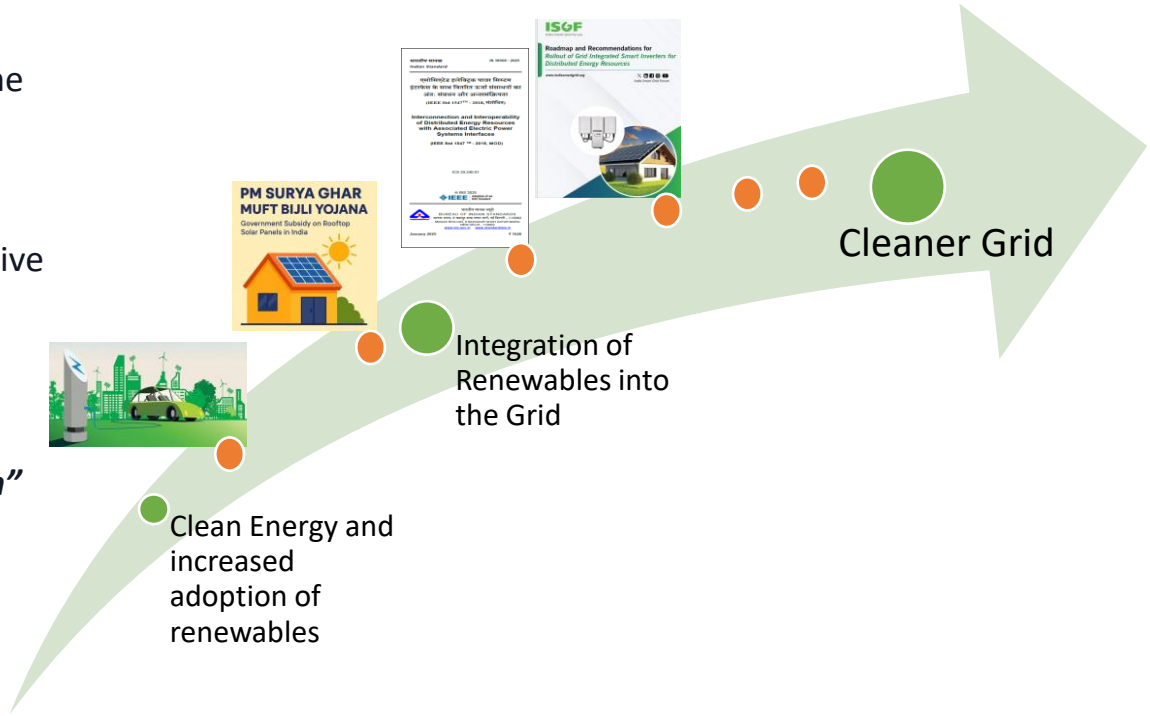
- It has identified **reverse power flow issues up to 66/11 kV substations**
- There are **68 number of 11kV feeders in Rajkot City of which 20+ feeders** are experiencing reverse power flow conditions



Why smart inverters are now a grid necessity

India is moving from a **synchronous grid** to **an inverter-dominated grid** at the distribution edge

- Rooftop solar, C&I DER, BESS and EV charging are growing fastest at LT/MV — the most operationally constrained part of the grid
- High inverter penetration can worsen voltage excursions, reverse power flow, protection miscoordination and power-quality issues unless DER is grid-supportive
- Smart inverters provide controllable functions: Volt-VAR, Volt-Watt, frequency droop, ride-through, ramp limits, and interoperability
- Policy objective: shift from ***"trip-and-disconnect" to "support-and-ride-through"*** while protecting consumers



Regulatory leverage : IS 18968:2025 (BIS) is India's national DER interconnection & interoperability standard
A modified adoption of IEEE 1547-2018 with National Annex J for Indian conditions

IS 18968:2025 — India's DER interconnection standard

What it governs: Performance categories (normal + abnormal), voltage regulation functions, ride-through, power quality, islanding, interoperability/communications, and test & verification

- Applies to DER at distribution voltage levels (LT/MV) and defines "Authority Governing Interconnection Requirements (AGIR)" roles
- National Annex J is normative (mandatory): adapts 60 Hz assumptions to India's 50 Hz grid, voltage tables, trip thresholds, volt-var settings, and enter-service criteria
- Current state: many advanced functions are DISABLED until enabled by the Area EPS operator — creating a clear governance pathway

National Annex J — India-specific modifications (selected)	
J- 1	50 Hz replaces 60 Hz for testing and operation
J-4/J-5	Enter-service tighter: 0.9–1.1 p.u.; 49.5–50.1 Hz; min delay 60
J-7/J-8	Reactive power capability: 60% injection & 60% absorption; wider VRef range 0.9–1.1 VN
J-10	Simplified voltage trip: V>1.1 pu (2 s), V<0.8 pu (2 s) — needs clarification vs category tables
J-11/J-12	Frequency: UF 47.5 Hz / OF 52 Hz (0.2 s default); continuous 47.5–52 Hz; droop defers to CEA specs

Key implementation point

Benefits materialize only when embedded into CEA/SERC interconnection processes, DISCOM agreements, and a national certification & lab-testing ecosystem

Performance categories

Assign the right requirement to the right DER Two dimensions: Normal (A/B) and Abnormal (I/II/III)

Normal Operating Performance: Category A or B (Reactive Power and Voltage Regulation)	Abnormal Operating Performance: Category I, II, or III (Ride-Through)
Category A: Minimum reactive power performance. Suitable for lower penetration areas and DER types with limited reactive power capability.	**Category I:** Essential ride-through meets basic bulk power system stability needs. Optional frequency droop for low frequency conditions. ROCOF ride-through 0.5 Hz/s.
Category B: Enhanced reactive power performance, all Category A capabilities PLUS watt-var (Active power, reactive power) mode and volt-watt (Voltage active power) mode. Mandatory for high penetration areas or areas with frequent large DER output variations.	**Category II:** Enhanced ride-through full BPS stability and coordination with existing reliability standards. Mandatory frequency droop for both low and high frequency. ROCOF: 2.0 Hz/s. **Category III:** Highest ride-through for very high DER penetration; addresses both BPS and distribution reliability. ROCOF: 3.0 Hz/s.

DER Type/Context	Normal Category	Abnormal Category	Rationale
Residential rooftop solar ≤ 10 kW (PM Surya Ghar)	Category A	Category I (Transitional) Category II (From 2027)	Low individual impact: phased upgrade as DISCOM DER capacity grows.
Commercial/Industrial rooftop 10 kW – 500 kW	Category B	Category II	Significant grid impact; must support volt-var and volt-watt functions.
Ground Mounted solar/wind > 500 kW at distribution level	Category B	Category II or III (per SLDC)	High penetration impact; SLDC coordination required per J-12.
Battery Energy Storage Systems (All sizes)	Category B	Category III	Full ride-through and dynamic voltage support capability; bidirectional operation.
EV charging infrastructure (Grid-interactive)	Category B	Category II or III	Bidirectional capability; growing role in frequency response.
Combined Heat and Power (CHP)/gas Gensets	Category A	Category I	Limited grid support capability: societal benefits justify lower category.
DER in high-penetration circuits (>15% solar peak penetration)	Category B (mandatory)	Category III (recommended)	Grid stability requires maximum reactive support and ride-through.

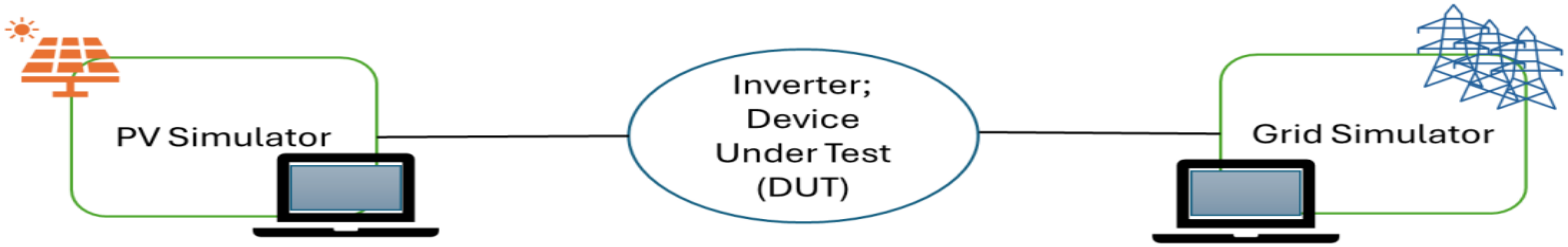
Recommended approach :

Adopt a national category assignment guideline (CERC/CEA) and allow SERCs to tighten by feeder penetration and criticality.

Testing of Inverter for IS-18968

Device under test: SolarEdge SE 8K (8 kVA, 3-phase)

- Phase 1: Basic Testing at UL Lab, Bengaluru
  - ISGF conducted the initial round
- Phase 2: Compliance Testing at NISE, Haryana
  - **Volt-Watt tests,**
  - **Volt-VAR tests,**
  - **Voltage Ride Through, and**
  - **Frequency Ride Through** were carried out

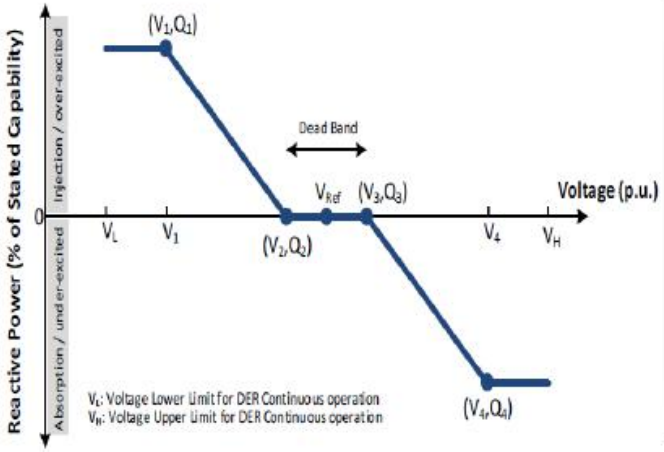


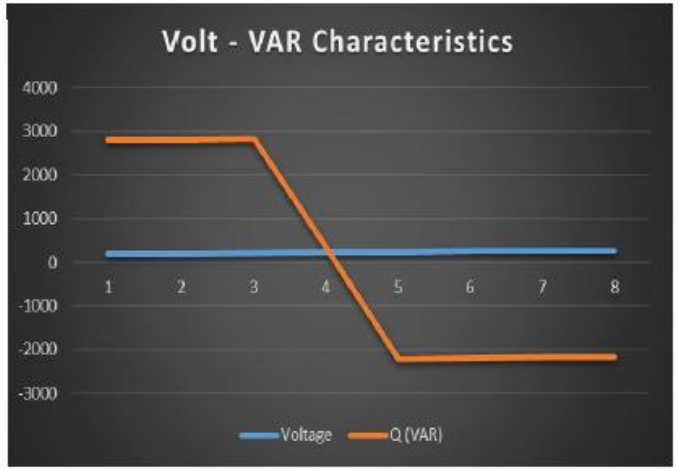
Volt - VAR Test

- The Volt–VAR test evaluates the capability of a smart PV inverter to autonomously regulate reactive power in response to deviations in grid voltage.
- This functionality plays a crucial role in stabilizing voltage conditions in modern distribution networks with high penetration of smart inverters.

OEM Set Limits

	P	V%	Q%
1	P0	50	-30
2	P1	92	-30
3	P2	96	0
4	P3	103	0
5	P4	107	30
6	P5	120	30





Lab Test Limits

	Voltage (V)	P (kW)	Q (VAR)	PF
1	184	6974	2614	0.93
2	190	7229	2616	0.94
3	207	7300	2629	0.94
4	230	7555	114	0.99
5	246	5549	-2461	0.95
6	250	6491	-2450	0.95
7	260	6552	-2439	0.93
8	265	5901	-2425	0.92

Injecting Reactive Power

Absorbing Reactive Power

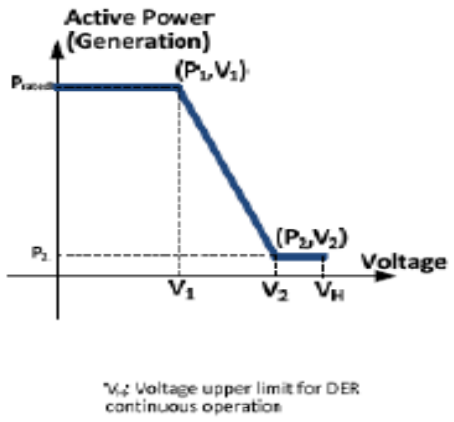
Test Result : Inverter injects and absorbs Reactive Power (Q) restricting the Active Power to 30%

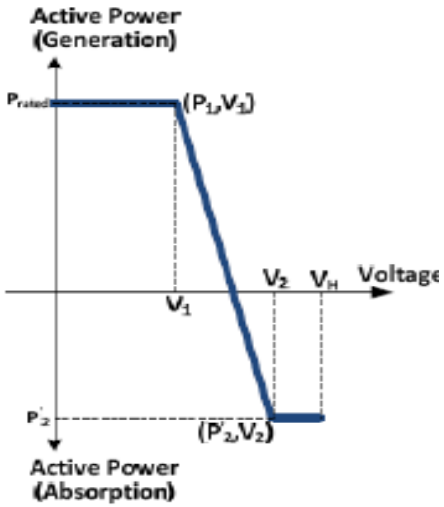
Volt - Watt Test

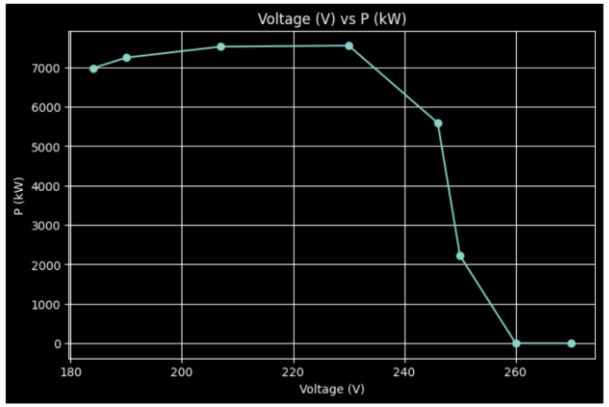
- The Volt–Watt test evaluates the capability of a smart PV inverter to autonomously curtail its active power output (P) in response to increasing grid voltage levels.
- The objective of this function is to prevent further voltage rise on distribution feeders, an increasingly common challenge in networks with high penetration of solar rooftop and distributed generation

OEM Set Limits

	P	V%	Q%
1	P0	0	100
2	P1	49	100
3	P2	100	100
4	P3	106	100
5	P4	110	0
6	P5	130	0







Lab Test Limits

	Voltage (V)	P (kW)	Q (VAR)	PF
1	184	6978	2553	0.53
2	190	7245	2615	0.93
3	207	7525	2601	0.94
4	230	7550	125	0.99
5	246	5589	-2438	0.92
6	250	2220	-2397	0.66
7	260	0.2	-702	0.04
8	270	-1.8	-775	0.03

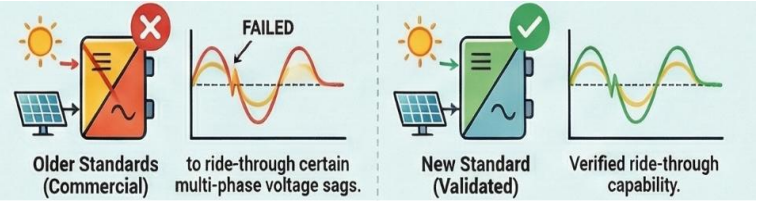
Test Result: Inverter injects Real power (P) but couldn't determine the absorption capability

Voltage Ride Through (VRT) Test

- The Voltage Ride Through (VRT) test is a mandatory evaluation to verify the ability of grid-connected smart inverters to remain connected and continue supporting the power system during short-duration voltage disturbances

Category I, II, III systems

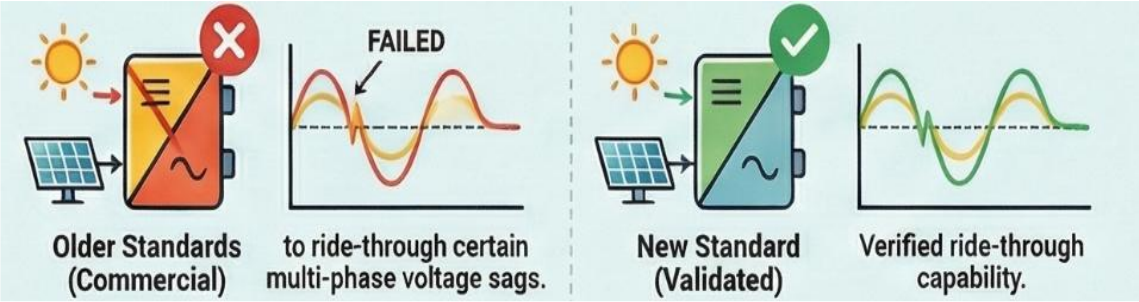
Voltage Range (p.u.)	Voltage Range (V)	Operating Mode/Response	Minimum Ride Through Time (Sec) Category I	Minimum ride-through time (s) Category II system	Minimum ride-through time (s) Category III system	Maximum Response Time (Sec)
V > 1.20	V > 276	Cease to Energize	NA	NA	NA	0.16
1.175 < V ≤ 1.20	270.25 < V ≤ 276	Permissive Operation	0.2	0.2	12	NA/NA/ 0.16
1.15 < V ≤ 1.175	264.5 < V ≤ 270.25	Permissive Operation	0.5	0.5	12	NA
1.10< V ≤ 1.15	253 < V ≤ 264.5	Permissive Operation	1	1	12	NA
0.88 ≤ V ≤ 1.10	202.4 ≤ V ≤ 253	Continuous Operation	Infinite	Infinite	Infinite	NA
0.70≤ V ≤ 0.88	161 ≤ V ≤ 202.4	Mandatory Operation	Linear Slope of 4S/230V Starting at 0.7S @ 161 V	Linear slope of 8.7 s/ 1 pu V starting at 0.3 s at 161 V	20	NA
0.50 ≤ V ≤ 0.88	115 ≤ V ≤ 202.4	Permissive Operation	0.16	0.32	10	NA
0.45 < V < 0.50	V < 115	Cease to Energize/ Permissive Operation	NA	0.32	1	NA/NA/ 0.083
0.30 < V < 0.45	V < 115	Cease to Energize/Permissive Operation	NA	0.16	NA	NA
V< 0.30	V <	Cease to Energize	NA	NA	NA	0.16



Test Result: As per the OEM-defined default settings, the inverter operates within the **LV limit of 184 V (0.8 p.u.) and the HV limit of 271 V (1.18 p.u.) after which the inverter trips and goes offline.**

Frequency Ride Through (FRT) Test

- The Frequency Ride Through (FRT) test is a mandatory evaluation to assess the capability of smart inverters to remain connected and support the grid during short-duration frequency deviations



Frequency Range (Hz)	Operating Mode	Minimum Time (S)
F > 51.6	No Ride Through Requirements apply to this Range	
51 < F ≤ 51.5	Mandatory Operation	299
49 ≤ F ≤ 51	Continuous Operation	Infinite
47.5 ≤ F ≤ 49	Mandatory Operation	299
F < 47.5	No Ride Through Requirements Apply to this Range	

Test Result: As per the OEM-defined settings, the inverter operates within the **LFRT limit of 48 Hz and the HFRT limit of 51.5 Hz.**

ISGF / UL / NISE Demo Results (2025)

Device under test: SolarEdge SE 8K (8 kVA, 3-phase)

- Phase 1: Basic Testing at UL Lab, Bengaluru
- Phase 2: Compliance Testing at NISE, Haryana(RT-LAB HIL, grid simulator)

What worked (capability confirmed)

- Volt-Watt: smooth curtailment from ~7.5 kW to 0 with rising voltage
- Volt-VAR: +2,800 VAR injection to −2,600 VAR absorption across voltage range

These functions are technically implementable with current-market inverters

What must change (settings & compliance)

- VRT/FRT: partial compliance — OEM default is different from standard
- OEM defaults differed from Annex J thresholds
- Need guidelines on implementation of the parameters.

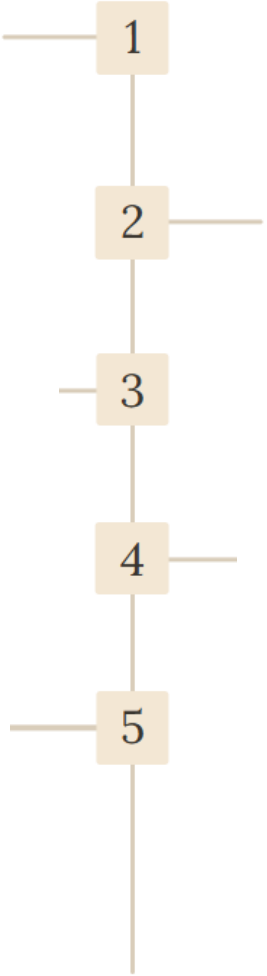
Test Result: OEM defaults differed from Annex J thresholds. Need guidelines on implementation of the parameters

IS 18968 Implementation Gaps and Path Forward

IS 18968 Implementation Gaps	
Limited Guidance on Grid Support Function Settings	•IS 18968 defines **grid support capabilities** (Volt-VAR, Volt-Watt, Frequency-Watt, etc.) but **does not define standardized default settings** for Indian grid conditions. •Lack of recommended **deadbands, droop slopes, and ramp rates** may lead to inconsistent behavior across utilities.
Limited Communication & Interoperability Requirements	•Communication protocols are mentioned but **not fully standardized for India**. •Lack of clear guidance on: • DER communication with **DERMS / SCADA** • Utility visibility and remote control • Interoperability across vendors.
Cybersecurity Framework Not Fully Defined	•Smart inverters introduce **grid-connected digital interfaces**. •IS 18968 lacks detailed requirements for: • Secure communications • Authentication • Utility command authorization.
Testing, Compliance and Testing Procedures Gaps	•Limited availability of **certified test labs in India** for smart inverter validation. •IS 18968 references performance functions but does not fully define: • Field verification procedures • Interoperability testing • Dynamic response validation.

Phase-wise Implementation of IS 18968:2025

2025–2032 Strategic Timeline



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